Applying Code-Switching Method for E-Learning in English and Japanese Vocabulary Acquisition Systems

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Abstract

Teaching vocabulary is an essential part of second language education. Until now many existing techniques tried to facilitate word acquisition. One method which has been paid less attention to is code-switching. In this paper we present an experimental system for computer assisted vocabulary learning in context using a code-switching based method, focusing on teaching Japanese vocabulary to foreign language learners. First, we briefly introduce our Co-Mix method for vocabulary teaching systems using code-switching phenomenon to support vocabulary acquisition. Next, we show how we utilize incidental learning technique with graded readers to facilitate vocabulary learning. We present the systems architecture, underlying technologies and the initial evaluation of the system's performance by using semantic differential scale. Finally, we discuss the evaluation results and further development of this technology.

Keywords: computer assisted e-learning, code-switching, teaching vocabulary, natural language processing

1. Introduction

In this day and age knowing foreign language is an essential skill necessary for cross-cultural communication. This increasing trend brings the problem of finding qualified teachers that can give students a chance to practice their language skills, and while coming across the right person is relatively easy in a big city, people living in rural or remote areas generally have less chance of finding qualified educators. The most essential part of second language (L2) learning is vocabulary (Ling, 2010). The problem of vocabulary acquisition is one of the most significant issues that have to be faced in order to master foreign languages. In a spoken language, about 1,800 words constitute about 80% of the spoken corpus (McCarthy, 2004). Because some frequent words are often repeated, it is thought that learners can understand a large proportion of foreign language conversation with a relatively small vocabulary (McCarten, 2007). The importance of vocabulary is explained by Wilkins, who states: "Without grammar very little can be conveyed, without vocabulary nothing can be conveyed." (Wilkins, 1972). In our research, the main scientific question we address is whether the code-switching phenomenon can expand students' vocabulary and enable them to understand words without given definitions.

2. Code-switching

In recent years, many approaches to vocabulary learning have been presented. New possibilities come with the code-switching phenomenon, in which a word from one language is used in a sentence in which the grammatical structure belongs to another language. Code-switching presents a chance for students to think about new foreign language words in a deeper manner and offers the potential for expansion of the second language vocabulary. Recently this technique has become a target of interest for researchers in different domains, like linguists (Coady & Huckin, 1997) and computer scientists (Labutov & Lipson, 2014). It can be defined as "the alternation of two languages within a single discourse, sentence or constituent" (Jamshidi & Navehebrahim, 2013). Generally, code-switching can be explained as a brief insertion of a word from one language into another. For example: "*The TENKI*[1] *forecast is great for the coming days.*" ('The WEATHER forecast is great for the coming days.') - Japanese word within an English sentence. There are also two possible locations where the switch can occur, i.e. intersentential (after the sentence) and intrasentential (within the sentence).

A majority of people using code-switching is bilingual and able to speak two languages. They can easily code-switch and use it to find better ways to convey different meanings. However, there is an increasing amount of evidence that code-switching can also be used by less proficient people to fill linguistic gaps whenever the learner of a second language encounters the problem of insufficient vocabulary (Nishimura, 1995). These two aspects create a common distinction between code-switching as an asset for bilingual people with a high competence in both languages, and as a "reparation tool for insufficiency in the second language" (Hamers & Blanc, 2000). In the last two decades of research, there seems to be an agreement on the positive effects of code-switching in learning and that it could be applied in second language learning to strengthen the learning outcome (Celik, 2003; Jamshidi, 2013; Lin, 2013). There is a common agreement that when code-switching is planned ahead of use, it can greatly contribute to more efficient understanding of a given topic. These effects of code-switching have been studied by Cook (2001) who also found that vocabulary learning can be facilitated by mixing two languages. Another interesting finding is demonstrated in the work presented by Lin (2013). According to this study, code-switching may increase the amount of cognitive processing by students. A larger cognitive effort is necessary to process new vocabulary,

[1] From here onward romanized transliterations of Japanese are italicised. English translations in parentheses accompany the transliterations when needed.

and students will learn new words more thoroughly. Although Lin's research did not show conclusive proof for higher effectiveness of code-switching for learning vocabulary, it indicated that code-switching does not affect the vocabulary acquisition in a negative way.

3. Related works

To the authors' best knowledge there exists no robust methodology or teaching application for code-switching based vocabulary acquisition in e-learning. The problem of automatic creation of study materials for English classes has been approached by Ginsburg (2012), who proposed a learning program to generate editable English lesson materials to speakers of Japanese. The author underlines that higher vocabulary skills are crucial in order to understand the words while using this application. He also provides reading exercises with highlighted lexical units and their dictionary definitions. We follow this general idea of an exercise generator by developing a new method for assessing the vocabulary proficiency level of the user. We also utilize an idea from a previous study by Mazur et al. (Mazur, Rzepka & Araki, 2012) regarding the method of increasing user engagement by introducing various language quizzes with the purpose of increasing users' motivation to study. The novelty of our method is found in combining the phenomenon of code-switching with several other existing methods of teaching vocabulary that have been proved to be effective, and implementing them in e-learning software to facilitate second language vocabulary acquisition.

4. Research background

The theoretical background of our research comes from many sources. We benefit from a study by Nation (2001) that brings an insight on how considerate amount of L2 vocabulary acquisition happens by incidental learning and long-term vocabulary growth is greater by incidental learning method. The vast majority of vocabulary is learned gradually through multiple exposures to a word over time. Another idea that contributed to our research originated in a study by Nagy Nagy & Herman (1985). This study claims that extensive reading should be promoted, because it can lead to greater vocabulary growth. Further research by Nagy et al. (Nagy, Anderson & Herman, 1987) also showed that even one or two exposures to a target word are able to contribute to a students' initial vocabulary knowledge. Krashen believes that language learners can learn new vocabulary and spelling more efficiently by receiving comprehensible input while they read, and postulated this concept in the Input Hypothesis (Krashen, 1985). Labutov & Lipson (2012) claim that humans tend to be good at inferring meaning and that single words can be understood from the context of the surrounding text in the paragraph. Further proof for the potential effectiveness of code-switching for language acquisition comes from the recent findings of Borovsky et al. (Borovsky, Elman & Kutas, 2012). This research contributes evidence that even a single exposure to a novel word in a constrained context results in the integration of the word to the reader's personal semantic base. It is contrary to a prior belief that a given word must be repeated with a sufficiently high frequency, and shows that words can be remembered incidentally while reading. Coady et al. (Coady & Huckin, 1997) have stated that many adult L2 learners are acquiring new words through reading to improve language proficiency.

5. Teaching vocabulary

5.1. Methodology

There are many existing methods of teaching second language vocabulary. Teaching vocabulary mostly involves making students remember new language units by presenting words many times before students can successfully memorize them. Repetition is one of the popular techniques, along with actively recalling a word to be learned. Coady & Hucking (1997) claim that acquiring a substantial vocabulary in the second language is necessary to attain competencies in other language skills, such as listening, speaking or writing. This research follows Krashen's belief that "language learners acquire vocabulary and spelling more efficiently by receiving comprehensive input while reading" (Krashen, 1985). Therefore, it can be asserted that new vocabulary units should not be presented only with their first language equivalents and without the presence of second language context. We follow this general idea with our graded readers approach (see subsection 5.2). Gibbons's research (Gibbons, 1987) indicates that nouns are the most often code-switched parts of speech. As many single lexical item switches are nouns, in our study we decided to utilize this finding in order to make the best use of the code-switching phenomenon. Further evidence of the importance of nouns is presented in the studies by Vihman (1998), who also proposed that nouns are more commonly used for the embedded language, because translations to the second language are often provided for them.

5.2. High frequency words

Our systems measure students' second language ability and adjust the difficulty of a given task to their level. They offer the opportunity for incidental learning using graded readers - texts graded according to high frequency word count, designed to give learners of a language practice in reading and used to support the extensive reading approach. Graded readers are comprehensible learning materials with the high frequency word count. These words are very important because they account for a large proportion of the content in written language. High frequency words are linguistically adjusted to learners' L2 competence and therefore can be used by students on different levels of proficiency. Knowledge of about 4,000 words usually covers 86.0% of a given text (Nation, 2001). Therefore, both teachers and learners should spend considerable time on their acquisition. We facilitate vocabulary acquisition through repetition of target high frequency words through a series of tests (quizzes) and presenting comprehensible examples of their usage. If there is a delay between the presentation of a word form and its meaning, students have an opportunity to make an effort to guess or recall its meaning. Nation (2001) states that guessing can only be successful if we provide a good clue to the meaning (incidental learning). Our method offers the chance to experience both incidental vocabulary acquisition and direct teaching - two popular ways of learning high frequency words.

6. CO-MiX Method

Co-Mix is a vocabulary teaching method for e-learning systems that harnesses the benefits of code-switching, incidental learning and advantages of learning vocabulary in context. Until now, we implemented it in two systems: Co-Mix English, for Japanese learners, and

currently presented Co-Mix Japanese (from now on also referred to as proposed system) aimed at foreigners who want to study Japanese. The modular architecture allows easy extensions of the system with additional functionalities. Our systems consists of two modules. The first one is responsible for the automatic generation of study materials based on given graded readers in L1 (Fig. 1). The other module, the Language Quiz Generator, checks the users' understanding of L2 vocabulary through a series of language tests. The system also responds to users' mistakes by providing them with example sentences that show the correct usage of a given word. Since each user possesses a different level of vocabulary knowledge, in the first step the Co-Mix systems evaluate user language competency (see subsection 6.2.) and present a study article appropriate to the user's vocabulary knowledge level. The text is used to introduce new vocabulary units. Both systems are provided with a selection of graded articles in Japanese for the Co-Mix English system, and their translations into English used by Co-Mix Japanese and baseline systems. The baseline system is a variation of our proposed system without the code-switching module for experimental purposes. It is equipped with some commonly used vocabulary learning techniques (see subsection 5.1.), however, it lacks the code-switching method presented in the proposed system. Another difference comes from the fact that instead of using readers in a mixed language with code-switching, the baseline system presents reading exercises entirely in L2 using the set of articles manually translated and proofread by a native speaker.

6.1. Study material generator

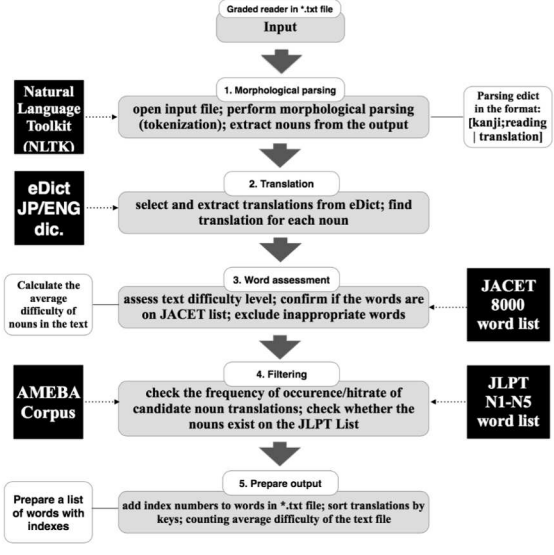


Fig. 1: Process of study material generator

The main function of this application is to automatically create editable learning materials from a given text file. The study material generator is fed with text files with graded readers in a given language and automatically generates an editable output file with tagged vocabulary items (nouns), their translations in second language for the proposed system, and their positions on the JACET 8000 vocabulary list (Uemura & Ishikawa, 2004). In case of the proposed system we use the JLPT vocabulary list[2]. Vocabulary lists are used to assess users' language ability and their prior knowledge. Our systems are using standardized lists of over 8,000 JLPT[3] words required for the exam across all 5 levels. Both lists provide the average difficulty of a text, which allows the system to categorize the text to an appropriate proficiency level.

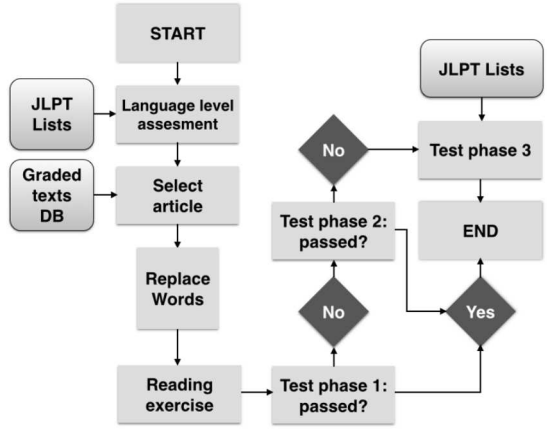


Fig. 2: Process of language quiz generator

Usually, preparing suitable reading materials for students presents a challenge for teachers. The study material generator can be used to manually create additional learning materials from given input texts. The process of the study material generator is as follows. First program searches the input file for a selected part of speech (e.g. noun) and checks if their English translations are available in the dictionary. Then the program analyzes the word rankings in a given reading and an average difficulty level is provided for the whole article. In the next step, the system generates an output text with selected words preceded by an index number and supplemented with a list of all chosen words, their proposed translations and their difficulty ranking. In this step, we allow the human teacher to review the result and eliminate possible mistakes that may be caused by limitations of the available list word coverage. This is a necessary step before handing over the materials to the language quiz generator and presenting them to users. A few search rules had to be established. In this study we focused nouns, which have an available translation in the dictionary, have a translation that consists of no more than three words, are on our frequency list and are not on the list of the first 500 most common words that students are most likely to know.

6.2. Language quiz generator

The language quiz generator utilizes reading materials prepared by the study material generator (Fig. 2). In the first step, it checks users' vocabulary level and provides them with reading exercises corresponding to their language level. It also generates quizzes to check the understanding of given vocabulary units after completing the selected reading exercise. The program

[2] The Japanese Language Proficiency Test, or JLPT, is a standardized test to evaluate and certify the Japanese language proficiency of non-native speakers."The JLPT is offered in five levels (N1, N2, N3, N4, N5). In order to measure Japanese-language proficiency as thoroughly as possible, test items are designed for each level."- Japanese Language Proficiency Test, accessed 1 October 2014, < https://www.jlpt.jp/e/about/points.html>.

[3] This selection is based on a database of about 18,000 vocabulary words that have appeared on the test with high frequency over the past 20 years.

works in the following manner: the user is presented with a set of 20 polar (yes/no) questions to determine the initial level of language proficiency. Next, the user is assigned to one of the five levels of language proficiency according to his/her final test score. We used the JLPT scale, which corresponds with the JACET 8000 vocabulary list (Japan Association of College English Teachers List of 8000 Basic Words) described by Uemura & Ishikawa (2004). Based on this selection, an appropriate text is selected and presented to the user in English with randomly selected nouns replaced with Japanese words. After finishing the reading exercise, users are presented with a set of a maximum of 20 test questions. The number of test questions depends on the user's language level, the difficulty of the selected reading, the number of available example sentences, and number of nouns extracted from the input file. The main purpose of these quizzes is to bring new vocabulary into productive use and enhance its memorization. The objective of each question is to type the word in second language. In the first phase of questions, random letters (not more than three and depending on word length) are presented to the user. If the user provides the system with correct answers, the test ends and automatic grading is provided. In the case of error, the system searches for an example sentence with the proper use of a given word and presents it to the user. We use sentences from the Tanaka corpus[4], a large public-domain collection of parallel Japanese and English sentence pairs. Next, mistaken words are repeated in the second phase of questions, where the difficulty level is slightly higher due to lack of hints and the necessity to write the target vocabulary in the sentence. Similarly to the second phase, the system repeats user mistakes in the third and final phase, where four words are presented and user has to select the correct answer.

7. Evaluation

7.1. Participants

Sixteen test subjects (university students and adults) participated in a preliminary evaluation experiment and interacted with the proposed system (Co-Mix Japanese). Among the respondents 12 were male and four were female. Ten participants were in their twenties and six participants were over 30 years old. In this experiment the target group were people who had some previous Japanese education, due to the fact that presented method requires basic Japanese skills, such as reading ability (hiragana, katakana and some basic Japanese kanji characters). Therefore, we decided to perform experiments with university students and adults with comparable language ability (the number of years both groups had studied foreign language was similar). We also selected 16 samples from baseline system experiment without code-switching to compare user attitudes towards baseline and proposed systems. Sixteen samples were selected to correspond with age, gender and education level of the proposed system users'. All participants were informed about the purpose of the study and the confidentiality of the results. Ten participants were adults (eight company employees, one self-employed and one housewife) and six were university students (undergraduate and graduate). Ten respondents (62.5%) have been learning Japanese from 2 to 5 years, three have been learning from 1 to 3 years, one have been studying less than 6 months, one have been studying less than 3 months and one have been studying more than 5 years. Seven respondents previously took the JLPT Exam (4 of them took Level 4, 2 of them level 5 and 1 of them Level 3). Thirteen participants also expressed their interest in taking JLPT exam in the future, two participants did not plan to take an exam, and one was not sure. Seven people were interested in taking Level 3, two participants wanted to start with Level 5, two participants were aiming at Level 2, one participant wanted to try Level 4 and one aimed at the highest Level 1.

7.2. Semantic Differential Scale

The participants of our experiment were asked to fill out a paper survey after they finished the learning tasks with a given system. In order to investigate users' attitudes towards the system, we used the semantic differential (SD) scale created by Osgood (1957). The SD scale utilizes polar words pairs such as "good - bad", "positive - negative" and "valuable - worthless" to measure the meaning of a given concept. The position marked 0 is labeled "neutral," the 1 position is labeled "slightly," the 2 position "quite," and the third position "extremely." In our experiment we decided to use 14 bipolar words and a seven-point scale to derive subjects' attitudes towards the given concepts and measure their opinions on a metrically controlled scale. The bipolar words were determined based on the most typical pairs of opposite expressions used to evaluate the peoples' attitude towards a computer system. We selected SD scale as a conceptual model of assessing the value of e-learning system because it fits in the framework to measure overall users perceived satisfaction with an e-learning system and learners behavior/usage of the system proposed by Levi (2006).

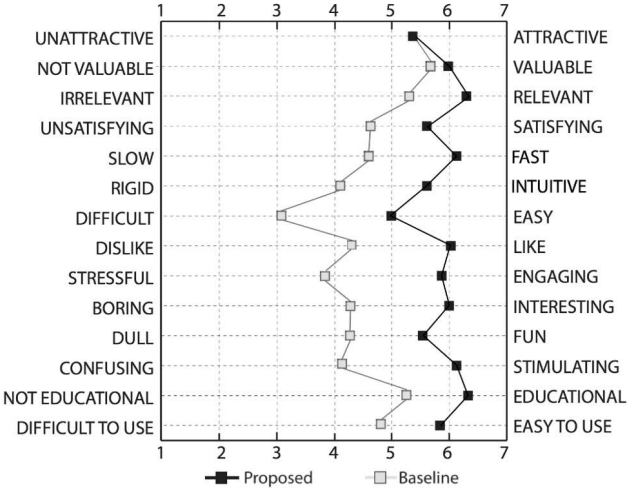


Fig.3: Semantic differential scale for proposed and baseline systems

7.3. Experiment results

The language level of each participant was determined by their number of correct answers in the initial vocabulary test. The average vocabulary level was 2.09 which corresponds to the Level 4, according to JLPT scale. In the test results only one person reached 100.0% correct answers in the first round, followed by 5 out of 15 in the second round (33.3%) and 7 out of 10 in the third round (70.0%). The average answering time for

[4] Tanaka Corpus of parallel Japanese-English sentences: http://www.edrdg.org/wiki/index.php/Tanaka_Corpus

the proposed system was 11.8 minutes. In case of the baseline system, 2 people out of 16 reached 100.0% correct answers in the first round (12.5%), followed by 4 out of 14 (28.5% of respondents) in the second round and 10 out of 10 (100.0%) reaching the maximum score in the third round. The average answering time for the baseline system was 8.8 minutes.

We collected the calculated average of system users' attitudes towards the Co-Mix Japanese system using the SD Scale. In Fig. 3., numbers closer to one represent the part of the spectrum assigned to the words on the left, whereas numbers close to seven are better characterized by the words on the right. After analyzing the results, we reached the conclusion that the average results of the proposed system were distinctly higher than the baseline. The overall average of the proposed system was 5.8 points, which was superior to the baseline system (4.5 points). The general trend line for the proposed system stayed within 5/6 at the higher end of the scale, whereas items in the baseline system such as "like/dislike" and " interesting/boring" diverted from the given trend line. On the other hand, three items ("stimulating/confusing", "educational/not educational", "easy to use/difficult to use") had similar results in both systems, which indicates general user satisfaction with both systems.

8. Conclusions and Future work

In this paper we presented Co-Mix, an experimental method for second language vocabulary acquisition in context using a code-switching based approach for studying Japanese vocabulary. We propose an innovative learning method that gives students a chance to expand their lexical knowledge. Further experiments confirmed the results of a previous study on a system for English vocabulary acquisition based on code-switching (to be published in near future). They demonstrated evidence that the proposed method is useful and effective way of expanding students' second language vocabulary. The high percentage of correct answers, making successful connections between meanings of words in two languages and positive comments from the participants suggest that a system for learning English vocabulary in context using a code-switching based approach has the potential to provide learners of Japanese with engaging learning activities. The participants improved their vocabulary scores after using the application, which demonstrates the pedagogical benefits of the Co-Mix method. The extension to support another language was successful which supports our belief that this method can be easily implemented in other languages with a proper morphological analyzer, dedicated study materials (graded readers) and high frequency word lists. Judging from the results, it can be asserted that user attitudes to our proposed system were positive and in most categories it significantly outperformed the baseline system. Additional rankings, based on the systems' performances and users' preferences, also indicated the proposed system as more favorable. High user satisfaction with our proposed system also provides motivation to improve and refine it in a number of different ways in the future.

We aim to incorporate additional functionalities into the Co-MiX systems and advance the project towards our initial idea to provide a chatterbot able to process user queries in mixed languages. Such Co-Mix systems enabled with a chatterbot could teach users second language vocabulary using the presented code-switching method in a human-like conversation. We are also planning to test our method on sentences with emotions expressed by emoticons. This would allow us to deal with shifting attitudes of users towards the system and respond to the problem of keeping users engaged in learning tasks and motivated to continue using the system in the future.

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